

case of diodes. Fig. 5 shows the characteristics curves for diodes 1N4001

to 1N4007.

It could also be variation of the cathode voltage vs cathode current as a function of temperature (Fig. 6) in the case of a thyristor, or dark current as a function of applied reverse bias in the case of a PIN photodiode, or transconductance versus drain current as a function of temperature in the case of power MOSFET. The curve in the first case emphasises that the relationship between the cathode voltage and cath-

ode current changes with temperature. The curve in the second illustration indicates critical dependence of the dark current of the PIN photodiode on the applied reverse bias voltage. And the third demonstrates significant reduction in the transconductance of the MOSFET at a fixed drain current with increase in temperature. These are all very important aspects of design and therefore should not be side-stepped.

Electrical characteristics. Electrical characteristics are usually given in tabular form indicating spread in parameter values that is likely to occur in a batch of devices in terms of minimum, typical and maximum values. It is a vital design input as in

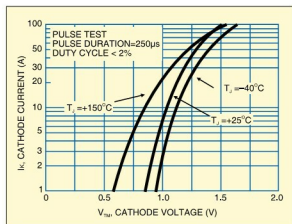


Fig. 6: Thyristor-cathode voltage vs. cathode current for different temperatures

Electrical Specifications $T_C = 25^\circ\text{C}$, Unless Otherwise Specified

PARAMETER	SYMBOL	TEST CONDITIONS		MIN	TYP	MAX	UNITS
Drain to Source Breakdown Voltage	BV_{DS}	$V_{GS} = 0V, I_D = 250\mu A$ (Figure 10)		200	-	-	V
Gate Threshold Voltage	$V_{GS(TH)}$	$V_{GS} = V_{DS}, I_D = 250\mu A$		2.0	-	4.0	V
Zero Gate Voltage Drain Current	I_{DSS}	$V_{DS} = \text{Rated } BV_{DS}, V_{GS} = 0V$ $V_{DS} = 0.8 \times \text{Rated } BV_{DS}, V_{GS} = 0V, T_J = 125^\circ C$		-	-	25	μA
On State Drain Current (Note 2)	$I_{D(ON)}$	$V_{DS} > I_{D(ON)} \times r_{DS(ON)MAX}, V_{GS} = 10V$		30	-	-	A
Gate to Source Leakage Current	I_{GSS}	$V_{GS} = \pm 20V$		-	-	± 100	nA
Drain to Source On Resistance (Note 2)	$r_{DS(ON)}$	$V_{GS} = 10V, I_D = 16A$ (Figures 8, 9)		-	0.07	0.085	Ω
Forward Transconductance (Note 2)	g_{fs}	$V_{DS} \geq 50V, I_D = 16V$ (Figure 12)		13	19	-	S
Turn-On Delay Time	$t_{D(ON)}$	$V_{DD} = 100V, I_D = 30A, R_G = 6.2\Omega, R_L = 3.2\Omega$ (Figures 17, 18) MOSFET Switching Times are Essentially Independent of Operating Temperature		-	20	30	ns
Rise Time	t_r			-	120	180	ns
Turn-Off Delay Time	$t_{D(OFF)}$			-	70	100	ns
Fall Time	t_f			-	80	120	ns
Total Gate Charge (Gate to Source + Gate to Drain)	$Q_{g(TOT)}$	$V_{GS} = 10V, I_D = 30A, V_{DS} = 0.8 \times \text{Rated } BV_{DS},$ $I_{D(REF)} = 1.5mA$ (Figures 14, 19, 20) Gate Charge is Essentially Independent of Operating Temperature		-	79	120	nC
Gate to Source Charge	Q_{gs}			-	13	-	nC
Gate to Drain "Miller" Charge	Q_{gd}			-	42	-	nC
Input Capacitance	C_{iss}	$V_{GS} = 0V, V_{DS} = 25V, f = 1.0MHz$ (Figure 11)		-	2000	-	pF
Output Capacitance	C_{oss}			-	800	-	pF
Reverse-Transfer Capacitance	C_{rss}			-	300	-	pF
Internal Drain Inductance	L_D	Measured between the Contact Screw on Header that is Closer to Source and Gate Pins and Center of Die	Modified MOSFET Symbol Showing the Internal Devices Inductances	-	5.0	-	nH
Internal Source Inductance	L_S	Measured from the Source Lead, 6mm (0.25in) from Header to Source Bonding Pad		-	12.5	-	nH
Thermal Resistance Junction to Case	$R_{\theta JC}$			-	-	0.83	$^\circ C/W$
Thermal Resistance Junction to Ambient	$R_{\theta JA}$	Free Air Operation		-	-	30	$^\circ C/W$

Fig. 7: Electrical characteristics of MOSFET type number IRF250